Mass Spectrometry Utility Suite

Project Section - User Manual

Scripts: [1] Project Folder Initializer | [2] Find Project | [3] Projects Overview

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1. Overview

The **Project** section of the Mass Spectrometry Utility Suite manages the lifecycle of proteomics projects - from creating the folder structure and recording metadata, to finding and reporting on existing projects.

All project data lives under a single configurable root directory (default: `Z:\Proteomics`). Projects are organised by analytics column, then by project name, and each project carries a `project_info.json` metadata file and is logged in a per-column `column_log.csv`.

Folder hierarchy

```
Z:\Proteomics\
  +-- Projects\
      +-- 2026-03-01_C20533039\   <-- analytics column folder
          |-- column_log.csv
          |-- column_info.json
          +-- 2026-03-09_MyProject\  <-- project folder
              |-- project_info.json
              +-- Plasma\
                  +-- Result\
              +-- pEV\
                  +-- Result\
```

> **Note**
> Both the column folder and the project folder are prefixed with the creation date (`YYYY-MM-DD_`) automatically. You never need to type the date prefix yourself.

Changing the root directory

If your proteomics data is not stored at `Z:\Proteomics`, open the main menu and navigate to **Settings** → **Set root directory**. Enter the new path and press `Enter`. The setting is saved and all scripts will use it from that point on.

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2. Script [1] - Project Folder Initializer

Creates a new project's folder structure and records its metadata. Can also be used to **update an existing project** (add sample folders, change PI, etc.).

How to launch

Select **[1] Project folder initializer** from the main menu and press `Enter` to confirm *Run*.

Step-by-step walkthrough

1. **Analytics Column ID**
   Type the column identifier exactly as printed on the column (e.g. `C20533039`).

   > **Important**
   > Do *not* include a date prefix. Enter only the column ID itself.

   If the column already exists in the system, existing projects under it are listed and the script notes that the column is already known.

2. **Analytics Column Description**
   An arrow-key menu appears showing previously used descriptions from the library.

   - `↑` / `↓` - Navigate the list
   - `Enter` - Select the highlighted description
   - `Del` - Remove the highlighted entry from the library
   - Select

     [+ Add new description]

     to type and save a new one

   Leave the new-description prompt blank to skip (description remains empty).

3. **New column details**
   (first use of a column only)
   If the column has not been seen before, you are asked for:

   - **First use date**

     - format `yyyy-MM-dd` (optional, press `Enter` to skip)

   This information is saved to `column_info.json` inside the column folder and is not asked again for the same column.

4. **Project Name**
   Type the project name. No date prefix required - it is added automatically.
   - Spaces are converted to underscores
   - Characters invalid in folder names are removed
   - Reserved names (e.g. *blank*, *prtc*, *sst*) are rejected

   If a project with this name already exists under the same column, the script switches to **update mode**
   (see below).

5. **PI Name**
   Enter the principal investigator's name. Leave blank to keep the current value (update mode) or leave unknown.

6. **Trap Column ID**
   (optional) Type the trap column identifier. Leave blank if no trap column is used.

7. **Trap Column Description**
   (only if a trap column was entered) Same arrow-key library menu as step 2, but for the trap column description library.

8. **Sample Subfolders**
   Enter comma-separated subfolder names (e.g. `Plasma,pEV,CSF`).
   - In update mode, leave blank to keep existing subfolders
   - Type new names to *add* them (existing ones are preserved)
   - Duplicates and reserved names are rejected with a clear message

   Each subfolder gets a `Result\` subdirectory inside it.

9. **Preview & Confirm**
   A summary tree of folders and metadata is shown. Use ↑ / ↓ to select *Yes, create folders* or *No, cancel*, then press `Enter`.

Updating an existing project

If you type a project name that already exists under the selected column, the script detects it automatically and shows:

```
Existing project found - leave fields blank to keep current values.
```

Each field prompt shows the current value in grey. Press `Enter` to keep it, or type a new value to overwrite. The **Project ID** and **project number** are always preserved from the original record. The CSV log row is updated in-place.

What the script creates

Item	Location	Description
Project folder	`Projects\<date_ColID>\<date_ProjectName>\`	Top-level project directory
Sample subfolders	Inside the project folder	One per sample type, each with a `Result\` sub-folder
`project_info.json`	Project folder root	Full project metadata (see field reference below)
`column_info.json`	Column folder root	Column metadata; created once per column
`column_log.csv`	Column folder root	One row per project; appended or updated in-place

project_info.json field reference

Field	Description
`ProjectID`	Random 8-character alphanumeric identifier, generated once
`Project`	Project name as entered (no date prefix)
`PI`	Principal investigator name
`AnalyticsColumn`	Analytics column ID
`ColumnDescription`	Analytics column description from the library
`TrapColumn`	Trap column ID (null if not used)
`TrapColumnDescription`	Trap column description (null if not used)
`ProjectNo`	Sequential project number within this column
`Created`	Creation timestamp (`yyyy-MM-dd HH:mm`)
`SampleFolders`	Array of all sample subfolder names

> **Tip - Reserved folder names**
> The following names cannot be used for project names or sample subfolders: `blank` , `raw_summary` , `prtc` , `sst` , `column_usage_history` . These are reserved for internal use.

3. Script [2] - Find Project

Searches all projects in the system and displays matching results. Supports searching by multiple fields and opening a project folder directly in Windows Explorer.

How to launch

Select **[2] Find project** from the main menu.

Searching

At the `Search` prompt, type one or more terms separated by commas or semicolons. The search is case-insensitive and partial matches are supported.

Searchable fields:

- Analytics column ID (e.g. `C20533039`)
- Project name
- PI name
- Project ID (the 8-character code)
- Trap column ID

Examples:

Input	Finds
`C20533039`	All projects on column C20533039
`Smith`	All projects where PI name contains "Smith"
`Plasma`	All projects named with "Plasma"
`C20533039, Smith`	Projects matching *either* term (OR logic)

Press `Enter` with a blank input to return to the main menu.

Results & navigation

Matching projects are listed with their ID, name, PI, column, and full folder path. After results are shown, a navigation menu appears:

Option	Action
Open in Explorer	Opens the project folder in Windows Explorer. If multiple results were found, a picker lets you choose which one to open.
New search	Clears the screen and prompts for a new search term
Back to main menu	Returns to the main menu
Exit	Closes the script

> **Tip**
> If the path shown in the results looks wrong, use **Settings** → **Set root directory** in the main menu to update the root path.

4. Script [3] - Projects Overview

Scans all projects in the system, applies optional filters, displays a summary grouped by column, and exports the results to an Excel file.

How to launch

Select **[3] Projects overview** from the main menu and confirm *Run*.

> **First run**
> On the first run (or if the **ImportExcel** PowerShell module is not installed), the script installs it automatically. This requires an internet connection and may take a moment.

Filters

After scanning, you are prompted for optional filters. Leave any field blank to use the default (all records).

Filter	Format	Blank default
Date from	`yyyy-MM-dd`	Oldest project creation date in the system
Date to	`yyyy-MM-dd`	Today's date
PI	Partial name, case-insensitive	All PIs
Column description	Arrow-key selection from library, or `Esc` to skip	All columns

The active filter range is always shown before the results, so you can confirm what date window is in use.

Console summary

Results are grouped by analytics column and displayed in the terminal:

```
-------------------------------------------------------
Column: C20533039 (3 project(s))
  #01  A8X3K2PL  Plasma_proteomics        Smith
  #02  B7R9W1QM  pEV_characterization     Lee
  #03  C4N2V6YT  CSF_cohort               Zhang
```

Export

The filtered results are automatically exported to an Excel file (`.xlsx`) in the `Projects` folder, named with a timestamp:

```
Excel : Z:\Proteomics\Projects\Projects_overview_20260309_143022.xlsx
```

The Excel file includes auto-filter headers, frozen top row, and the following columns:

Column	Description
Column	Analytics column ID
ColumnDescription	Column description from library
ProjectNo	Sequential project number (zero-padded)
ProjectID	Unique 8-character project identifier
Project	Project name
PI	Principal investigator
TrapColumn	Trap column ID
TrapColumnDescription	Trap column description
Created	Project creation timestamp
SampleCount	Number of sample subfolders
SampleFolders	Sample subfolder names (semicolon-separated)

After export, select **Open file** in the navigation menu to open the Excel file immediately.

5. Folder & File Structure Reference

Path	Created by	Purpose
`Projects\<date_ColID>\`	Script [1]	One folder per analytics column; named with date of first use and column ID
`...\column_info.json`	Script [1]	Column metadata: ID, description, first use date
`...\column_log.csv`	Script [1]	One row per project on this column; updated each time script [1] runs
`...\<date_ProjectName>\`	Script [1]	One folder per project; date-prefixed at creation
`...\project_info.json`	Script [1]	Full project metadata; source of truth for scripts [2] and [3]
`...\<SampleFolder>\Result\`	Script [1]	Per-sample result directory for DIA-NN output and reports
`Projects\Projects_overview_*.xlsx`	Script [3]	Filtered project export; timestamped filename